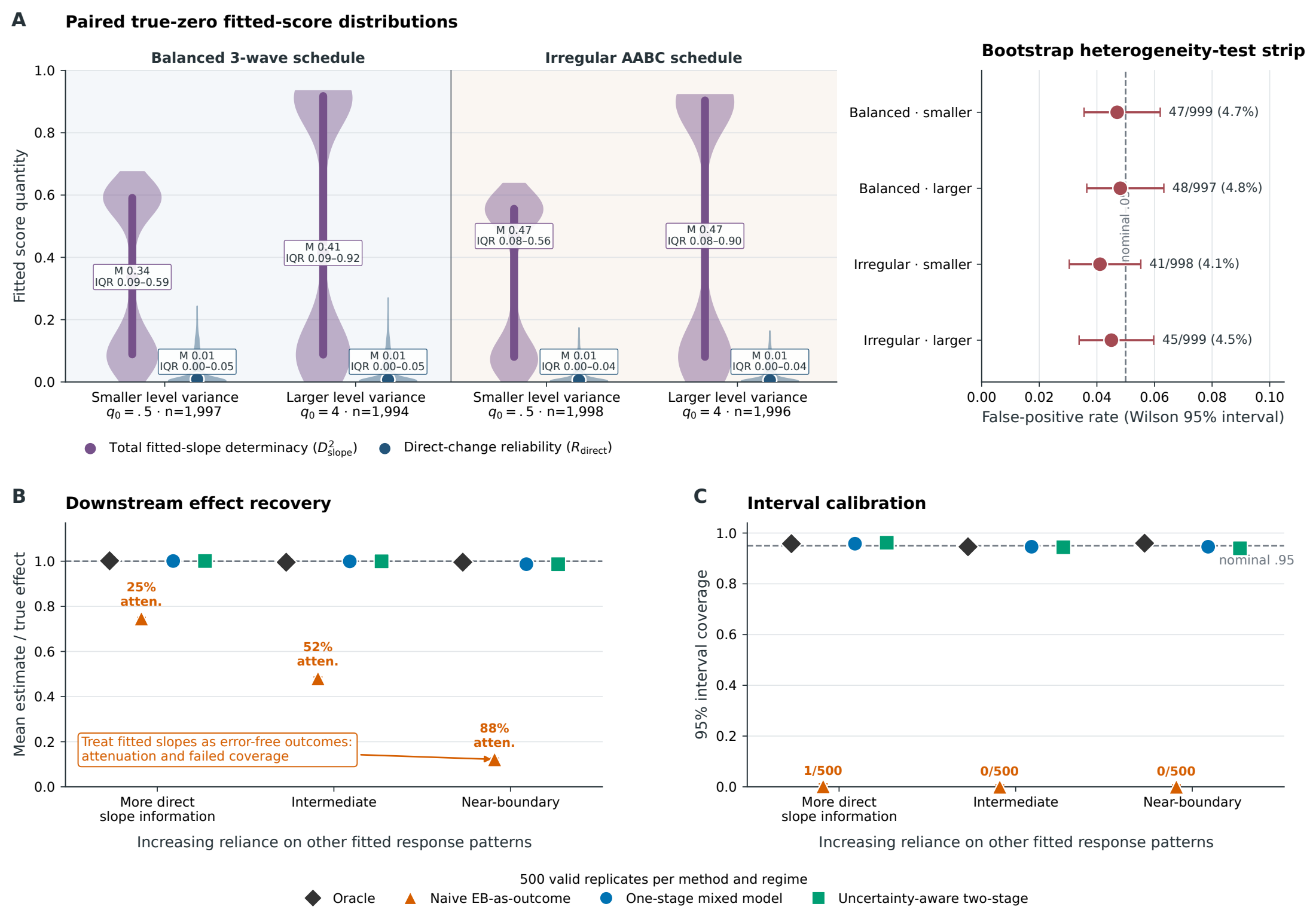

A
Paired true-zero fitted-score distributions
Balanced 3-wave schedule
Irregular AABC schedule
Fitted score quantity
1.0
0.8
0.6
0.4
0.2
0.0
M 0.34
IQR 0.09–0.59
M 0.01
IQR 0.00–0.05
M 0.41
IQR 0.09–0.92
M 0.01
IQR 0.00–0.05
M 0.47
IQR 0.08–0.56
M 0.01
IQR 0.00–0.04
M 0.47
IQR 0.08–0.90
M 0.01
IQR 0.00–0.04
Smaller level variance
$q_0 = .5$ · n=1,997
Larger level variance
$q_0 = 4$ · n=1,994
Smaller level variance
$q_0 = .5$ · n=1,998
Larger level variance
$q_0 = 4$ · n=1,996
Total fitted-slope determinacy (D^2_{slope})
Direct-change reliability (R_{direct})
Bootstrap heterogeneity-test strip
Balanced · smaller
Balanced · larger
Irregular · smaller
Irregular · larger
47/999 (4.7%)
48/997 (4.8%)
41/998 (4.1%)
45/999 (4.5%)
nominal .05
0.00
0.02
0.04
0.06
0.08
0.10
False-positive rate (Wilson 95% interval)
B
Downstream effect recovery
Mean estimate / true effect
1.0
0.8
0.6
0.4
0.2
0.0
25%
atten.
52%
atten.
88%
atten.
Treat fitted slopes as error-free outcomes:
attenuation and failed coverage
More direct
slope information
Intermediate
Near-boundary
Increasing reliance on other fitted response patterns
C
Interval calibration
95% interval coverage
1.0
0.8
0.6
0.4
0.2
0.0
nominal .95
1/500
0/500
0/500
More direct
slope information
Intermediate
Near-boundary
Increasing reliance on other fitted response patterns
500 valid replicates per method and regime
Oracle
Naive EB-as-outcome
One-stage mixed model
Uncertainty-aware two-stage